

Reasoning report: Wild type only: iPSC to T lineage, day 40

Loop example-ipsc-tcell-wt · request ipsc-tcell-wt-d40-001 · 4 decisions · 3 of 30 iterations · ended in **protocol_succeeded**

What this report is. It records why this loop made each choice, with the references and rationales behind them. The bioreactor was a **synthetic stand-in**, so nothing here is biological evidence, and the decisions were authored by deterministic policy (not a model). Read a recovered parameter as evidence that the search works, not as a finding about any gene or any cell.

Contents

- [The question and the envelope](#)
- [Outcome](#)
- [How the search moved](#)
- [Every decision, with its reasoning](#)
- [The protocol it ended on](#)
- [References](#)
- [Limitations](#)

The question and the envelope

Autonomy	autonomous
Reactor	synthetic_standin
Stand-in	ipsc_tcell
Iteration budget	30

Outcome

Final verdict **SUCCESS**: SUCCESS: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; QC PASS.



Final target_cells_per_input_cell per arm, in cells/input_cell, against the required value. Bars are the best configuration each arm ended on.

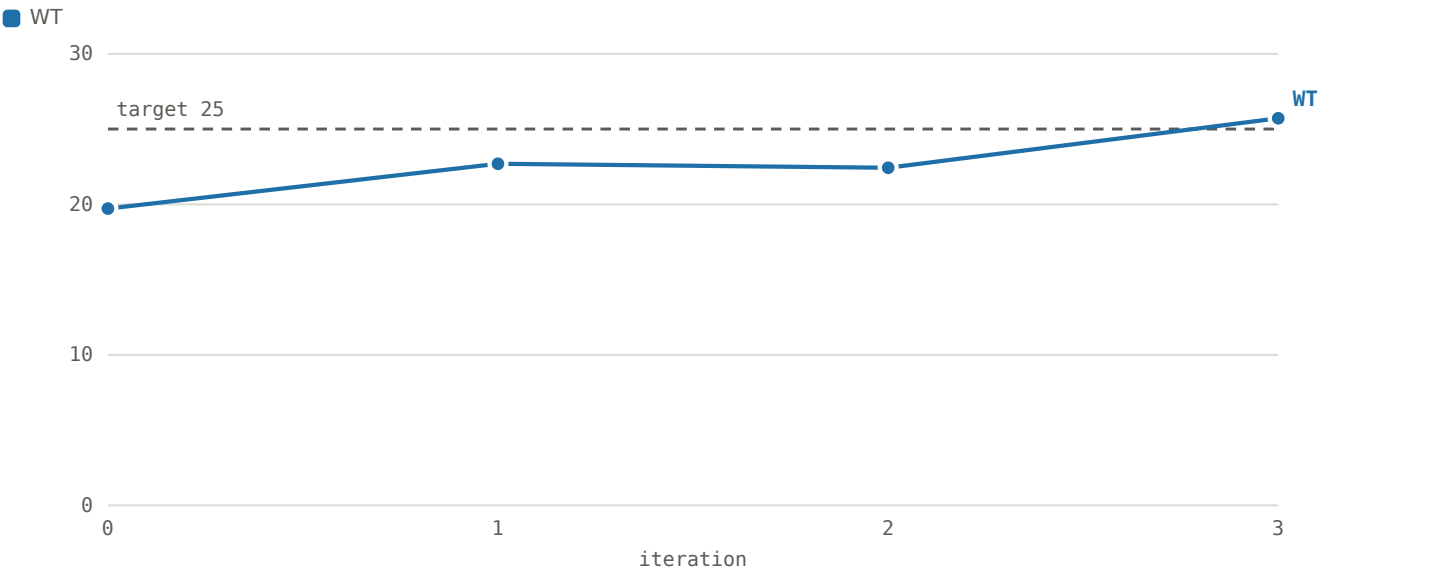
ARM	OBSERVED	REQUIRED	TARGET	QC
WT	25.718	25	MET	all pass

Best reading reached per arm

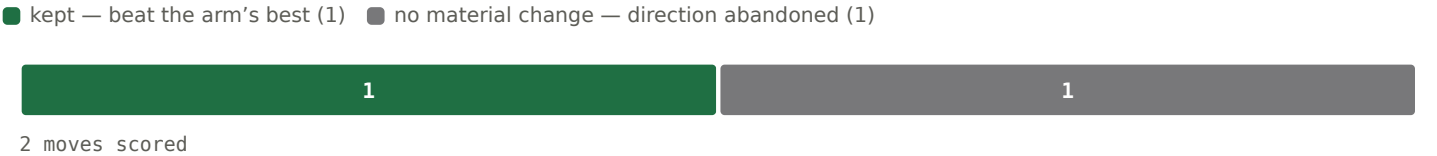
ARM	BEST	ITERATION	PROTOCOL
WT	25.718	3	ipsc-tcell-wt-d40-001-it3

How the search moved

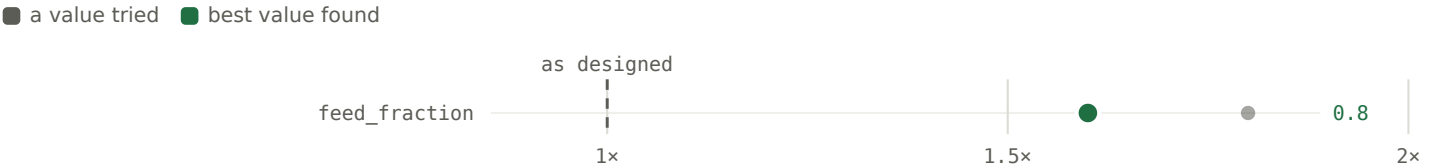
The optimiser is a bounded coordinate search with backtracking. It moves one lever per arm per iteration, scores it against that arm's best reading so far, and reverts anything that does not beat it. It has no model of the process and cannot see interactions between levers.



One point per stand-in run. The optimiser moves one lever per arm per iteration and reverts any move that does not beat that arm's best reading, which is why a line can step back and then recover. The dashed rule is the target the request asked for.



Every move the search made, by what the next run said about it. A reverted move is not a failure of the search — it is the search refusing to keep a change that did not pay, which is what stops it drifting away from a good configuration.



Each row is one lever, with every value the search tried on it, as a multiple of what the designed protocol started with. A shared absolute axis would be meaningless because the levers carry different units. The filled marker is the value that gave that lever its best reading, printed in its own units on the right.

Reading per iteration

ITERATION	WT
0	19.715

ITERATION	WT
1	22.694
2	22.422
3	25.718

Best reading per arm

ARM	BEST	AT ITERATION
WT	25.718	3

Every lever the search touched

PARAMETER	SCOPE	STEP	KIND	VALUES TRIED	EXHAUSTED	BEST
feed_fraction	all arms	culture:feed_fraction	culture	0.8, 0.9	up	0.8

Change log, iteration by iteration

INTO ITERATION	STEP	CHANGE	ADDRESSES
1	culture:feed_fraction	feed_fraction: 0.5 -> 0.8 fraction (all arms)	brief-ipsc-tcell-wt-d40-001-it0
2	culture:feed_fraction	feed_fraction: 0.8 -> 0.9 fraction (all arms)	brief-ipsc-tcell-wt-d40-001-it1
3	S23	IL-7: 10 -> 16 ng/mL (all arms)	brief-ipsc-tcell-wt-d40-001-it2

Every decision, with its reasoning

1. revise_protocol — decision-00, iteration 0

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 0 protocol ipsc-tcell-wt-d40-001-it0 did not work: target:WT NOT_MET (observed 19.7151, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 19.7151 vs required 25.0 (shortfall 21%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-wt-d40-001-it0

Iteration 0 protocol ipsc-tcell-wt-d40-001-it0 did not work: target:WT NOT_MET (observed 19.7151, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:WT **NOT_MET** — observed 19.715, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

2. revise_protocol — decision-01, iteration 1

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 1 protocol ipsc-tcell-wt-d40-001-it1 did not work: target:WT NOT_MET (observed 22.6938, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

ACTION	WHY IT WAS NOT CHOSEN
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 22.6938 vs required 25.0 (shortfall 9%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-wt-d40-001-it1

Iteration 1 protocol ipsc-tcell-wt-d40-001-it1 did not work: target:WT NOT_MET (observed 22.6938, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:WT **NOT_MET** — observed 22.694, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage

PARAMETER	DIRECTION	STEP	BASIS
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

3. revise_protocol — decision-02, iteration 2

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 2 protocol ipsc-tcell-wt-d40-001-it2 did not work: target:WT NOT_MET (observed 22.4218, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

ACTION	WHY IT WAS NOT CHOSEN
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 22.4218 vs required 25.0 (shortfall 10%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-wt-d40-001-it2

Iteration 2 protocol ipsc-tcell-wt-d40-001-it2 did not work: target:WT NOT_MET (observed 22.4218, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:WT **NOT_MET** — observed 22.422, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage

PARAMETER	DIRECTION	STEP	BASIS
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

4. protocol_succeeded — decision-03, iteration 3

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	no
Routed to	human
Validated against	protocol_succeeded, escalate_to_human

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict SUCCESS: SUCCESS: target target_cells_per_input_cell $\geq$ 25 cells/input_cell at day 40 met by ['WT']; QC PASS. The envelope allowed ['escalate_to_human', 'protocol_succeeded'] and the policy rule for this verdict selects 'protocol_succeeded'. Every arm that the request holds to the target met it, and every QC criterion passed. There is nothing further to gather evidence for and nothing left to revise, so the loop reports success. This is not a released process: confirmation runs and human QA sign-off are outside this loop.

Stated reason

All target arms met the desired output and every QC criterion passed. SUCCESS: target target_cells_per_input_cell $\geq$ 25 cells/input_cell at day 40 met by ['WT']; QC PASS. Release still requires human QA sign-off and confirmation runs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
escalate_to_human	Allowed for this verdict but not selected: the policy rule for SUCCESS prefers 'protocol_succeeded'. Every arm that the request holds to the target met it, and every QC criterion passed. There is nothing further to gather evidence for and nothing left to revise, so the loop reports success. This is not a released process: confirmation runs and human QA sign-off are outside this loop.
stop_budget	Refused by the envelope: The target was met; stopping on budget would misreport the outcome.

The protocol it ended on

This is the configuration the search held at the end, which under its own accept/reject rule is the best it found. Read the provenance column before reading any number: a design choice is a value the loop chose, not a value anyone measured.

29 quantities

Every quantity in the final protocol by where its value came from. A bar that is entirely design choice means nothing in the recipe is attributable to a publication, which is a statement about the evidence available, not about the recipe being wrong.

Protocol	ipsc-tcell-wt-d40-001-it3
Title	Designed iPSC -> T cell schedule to day 40 (WT) (iteration 3: IL-7 increase)
Day 0 is	iPSC seeding into the vessel
Approved by	automated stand-in run (no human approved this; no wet-lab run is authorised by this protocol)
Quantities	Design choice 29

Every quantity, with its provenance

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
culture_system.working_volume	100	mL	Design choice	–	Working volume chosen here for a bench-scale run. Not a retrieved value.
culture_system.parameters.dissolved_oxygen	21	%	Design choice	–	Atmospheric oxygen, chosen here; a hypoxic early phase is a plausible alternative this schedule does not test.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
culture_system.parameters.feed_fraction	0.9	fraction	Design choice	–	Search step from the search itself, after scoring the previous move. Lever: 'feeding regime (feed_fraction)' increase. Basis given for the lever: Continuing an ascent: the previous step on this lever (0.5 -> 0.8 fraction) improved WT, so the same direction is tried again. Original basis: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. Explored because the implicated levers are exhausted. A fed-batch split changes how much medium the culture gets and therefore the total yield, so a gain here says something about the vessel, not about the biology.. Moved from 0.8 to 0.9 fraction for every arm. Brief: brief-ipsc-tcell-wt-d40-001-it1. This value came from a bounded search step, not from any publication, so it is a design choice.
culture_system.parameters.temperature	37	degC	Design choice	–	Standard culture temperature.
mesoderm.medium	serum-free, xeno-free basal medium with mesoderm supplements	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
mesoderm.S01 (seed)	200,000	cells/mL	Design choice	–	Starting iPSC density chosen here as a mid-range aggregate seeding density. No retrieved value supports it.
mesoderm.S02 (BMP4)	20	ng/mL	Design choice	–	Standard mesoderm-inducing signal in pluripotent-stem-cell haematopoietic differentiation. Dose is a mid-range starting point chosen here, not a retrieved value.
mesoderm.S03 (VEGF)	50	ng/mL	Design choice	L02	Supports endothelial specification and the endothelial-to-haematopoietic transition. Dose chosen here.
mesoderm.S04 (FGF2)	20	ng/mL	Design choice	–	General proliferative support across the early stages. Dose chosen here.
mesoderm.S05 (CHIR99021)	4	uM	Design choice	–	GSK3 inhibitor used as a Wnt agonist for mesoderm patterning, applied as a short pulse. Dose and pulse length chosen here.
hemogenic_endothelium.medium	serum-free basal medium with haematopoietic supplements	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.
hemogenic_endothelium.S07 (VEGF)	50	ng/mL	Design choice	L02	Supports endothelial specification and the endothelial-to-haematopoietic transition. Dose chosen here.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
hemogenic_endothelium.S08 (FGF2)	20	ng/mL	Design choice	–	General proliferative support across the early stages. Dose chosen here.
hemogenic_endothelium.S09 (SCF)	50	ng/mL	Design choice	–	Haematopoietic progenitor survival and proliferation factor. Dose chosen here.
hemogenic_endothelium.S10 (SB431542)	5	uM	Design choice	–	TGF-beta receptor inhibitor, applied briefly to favour the haematopoietic transition. Dose and window chosen here.
t_commitment.medium	serum-free lymphoid medium on a DLL4-presenting surface	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.
t_commitment.S12 (DLL4)	500	ng/mL	Design choice	L01	Notch ligand presentation is the lever that commits progenitors to the T lineage; the cited work reports that providing DLL4 increases T-competent progenitor output. The direction is cited; the surface density here is a design choice.
t_commitment.S13 (SCF)	50	ng/mL	Design choice	–	Haematopoietic progenitor survival and proliferation factor. Dose chosen here.
t_commitment.S14 (FLT3L)	10	ng/mL	Design choice	–	Supports lymphoid progenitor expansion alongside SCF. Dose chosen here.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
t_commitment.S15 (IL-7)	10	ng/mL	Design choice	L04	The central homeostatic cytokine for T-lineage survival and proliferation, and the lever the cited BACH2 work implicates for maintaining a less differentiated state. Dose chosen here.
t_commitment.S16 (TP0)	20	ng/mL	Design choice	–	Supports early haematopoietic progenitors; withdrawn mid-stage. Dose and window chosen here.
maturation.medium	serum-free lymphoid medium	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.
maturation.S18 (anti-CD3)	50	ng/mL	Design choice	L03	TCR/CD3 agonist providing the maturation signal, applied as a limited pulse because sustained TCR pressure drives differentiation. Dose and pulse length chosen here.
maturation.S19 (IL-7)	10	ng/mL	Design choice	L04	The central homeostatic cytokine for T-lineage survival and proliferation, and the lever the cited BACH2 work implicates for maintaining a less differentiated state. Dose chosen here.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
maturation.S20 (SCF)	50	ng/mL	Design choice	–	Haematopoietic progenitor survival and proliferation factor. Dose chosen here.
expansion.medium	serum-free lymphoid medium	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.
expansion.S23 (IL-7)	16	ng/mL	Design choice	–	Search step from unimplicated_search. Lever: 'IL-7 (expansion)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 10 to 16 ng/mL for every arm. Brief: brief-ipsc-tcell-wt-d40-001-it2. This value came from a bounded search step, not from any publication, so it is a design choice.
expansion.S24 (IL-15)	10	ng/mL	Design choice	L03	Homeostatic cytokine supporting expansion with less differentiation pressure than IL-2. Dose chosen here.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
expansion.S25 (IL-2)	50	ng/mL	Design choice	–	Classical T-cell expansion cytokine. Included for expansion drive; dose chosen here.

References

Publications

1. Fine-tuning BACH2 dosage balances stemness and effector function to enhance antitumor T cell therapy. Nat Immunol. 2025. [10.1038/s41590-025-02389-z](https://doi.org/10.1038/s41590-025-02389-z)
full text not retrieved
Used for:

- claim L05 (bach2_dosage_effect)

2. Michaels YS, Edgar JM, Major MC, et al. DLL4 and VCAM1 enhance the emergence of T cell-competent hematopoietic progenitors from human pluripotent stem cells. Sci Adv. 2022;8(34):eabn5522. [10.1126/sciadv.abn5522](https://doi.org/10.1126/sciadv.abn5522) · PMC PMC9401626
full text not retrieved
Used for:

- claim L01 (notch_ligand_requirement)
- claim L02 (stage_order)

3. Roychoudhuri R, Clever D, Li P, et al. BACH2 regulates CD8+ T cell differentiation by controlling access of AP-1 factors to enhancers. Nat Immunol. 2016;17(7):851-860. [10.1038/ni.3441](https://doi.org/10.1038/ni.3441) · PMC PMC4918801
full text not retrieved
Used for:

- claim L03 (bach2_terminal_differentiation)

4. Tsukumo S, Unno M, Muto A, et al. Bach2 maintains T cells in a naive state by suppressing effector memory-related genes. Proc Natl Acad Sci USA. 2013;110(26):10735-10740. [10.1073/pnas.1306691110](https://doi.org/10.1073/pnas.1306691110) · PMC PMC3696756
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Used for:

- claim L04 (bach2_naive_maintenance)

Limitations

- The reactor was a synthetic stand-in, so no number in this report is a measurement of any real cell population. The stand-in is phenomenological and is not a validated digital twin.
- Every decision here was authored by the deterministic policy. The reasoning shown is a description of a rule in the code, not a model's reasoning. No LLM ran in this loop.

- Not one quantity in the final protocol is attributed to a publication: all 29 of them are design choices. The citations support the direction of a lever and nothing numeric.
- 1 replicate per arm. Every between-arm difference is directional only, and the search was climbing a surface it could not distinguish from noise below its own step size.
- A loop that reaches a target has not produced a validated or released process. Confirmation runs, a replicate design and human QA sign-off are all outside it.
- A designed schedule is not an optimised or validated schedule.
- Designed, not extracted. No quantity is attributed to a publication.
- Revised by the deterministic coordinate search in biosense.production.revise. Every revised quantity is a design choice, and no searched value is attributed to a publication.

Generated by biosense.production.report from the JSON in example-ipsc-tcell-wt. Nothing in this report was written by hand, and nothing in it is clinical or manufacturing guidance.